

Submission A solution to Problem 3

Problem statement and interpretation

I interpret the question in the universal sense: determine those $p \in [0, 1]$ for which

$$\Pr \left[\sum_{i=1}^m w_i v_i \geq p \right] \geq p \quad (\text{P})$$

holds for every $m \geq 1$ and every non-negative probability vector $(w_1, \dots, w_m)$, with v_i i.i.d. Bernoulli(p).

The counterexamples in Proposition 4 show that no value outside

$$[0, 1/3] \cup \{1/2, 1\}$$

can work. Conversely, a full classification would follow from the small- p assertion (Conjecture 1) that all $0 \leq p \leq 1/3$ work. That assertion remains the only essential gap in this draft. What is proved unconditionally below is the complete negative direction, several infinite positive families, closure properties, equivalent forms of the remaining assertion, and a coalescence route that is ruled out by explicit examples.

Positive values proved here

Let $\mathcal{G}$ be the set of probabilities for which (P) holds universally.

Proposition 1 (a coloring criterion). *Let $p = a/b$, where $1 \leq a \leq b$ are integers. Suppose that every non-negative b -tuple $x_1, \dots, x_b$ with $\sum_j x_j = 1$ has at least*

$$\frac{a}{b} \binom{b}{a} = \binom{b-1}{a-1}$$

a -subsets $J \subseteq [b]$ satisfying $\sum_{j \in J} x_j \geq a/b$. Then $a/b \in \mathcal{G}$.

Proof. Randomly color each original index i independently and uniformly with one of b colors, and let X_j be the total weight of color j . For every coloring outcome, the hypothesis

gives at least $\binom{b-1}{a-1}$ good a -subsets of colors. Hence, for a uniformly random a -subset J of colors,

$$\Pr \left[\sum_{j \in J} X_j \geq a/b \right] \geq \frac{\binom{b-1}{a-1}}{\binom{b}{a}} = \frac{a}{b}.$$

By symmetry the same lower bound holds for any fixed a -subset J . For such a fixed J , membership of an original index in J is an independent Bernoulli(a/b) event, which is exactly (P). $\square$

The criterion immediately gives the reciprocal points $p = 1/k$, $k \geq 3$: among k color classes at least one has weight at least $1/k$.

It also gives the following second family.

Lemma 1. *Let $b \geq 6$, and let $x_1, \dots, x_b \geq 0$ with $\sum_i x_i = 1$. Then at least $b - 1$ unordered pairs $\{i, j\}$ satisfy*

$$x_i + x_j \geq \frac{2}{b}.$$

Proof. Put $z_i = bx_i/2$. Then $\sum_i z_i = b/2$, and we must find at least $b - 1$ pairs with $z_i + z_j \geq 1$.

Maintain a remaining set R of size s and sum at least $s/2$. Let u and a be the smallest and largest elements of R . Since the average of the elements of R is at least $1/2$, its largest element satisfies $a \geq 1/2$. If $u + a \geq 1$, then a makes good pairs with all other $s - 1$ elements and we stop. Otherwise $u + a < 1$. Let h be the number of other elements making a good pair with a . If $h \leq (s - 2)/2$, then the h partners of a are at most a , while the other $s - 1 - h$ elements are $< 1 - a$, so

$$\sum_{z \in R} z < a + ha + (s - 1 - h)(1 - a) = s - 1 - h + a(2h - s + 2) \leq s/2.$$

The last inequality uses $a \geq 1/2$ and $2h - s + 2 \leq 0$. This contradiction shows that $h \geq \lfloor s/2 \rfloor$. Count these h pairs, delete u and a , and continue. The remaining sum is $> s/2 - 1 = (s - 2)/2$, so the invariant is preserved. Pairs counted at different stages are distinct. If the process reaches two remaining elements, their sum is at least 1 by the invariant, so this final pair is counted and the process stops; if one element remains, no further pair is available. These terminal possibilities are included in the “never stops early” count below.

If the procedure never stops before the terminal case, then for $b = 2r$ it counts $r + (r - 1) + \dots + 1 \geq 2r - 1 = b - 1$, and for $b = 2r + 1$ it counts $r + (r - 1) + \dots + 1 \geq 2r = b - 1$, using $r \geq 3$.

If it stops after t deletion steps, then $s = b - 2t$ elements remain and the number counted is

$$\sum_{i=0}^{t-1} \left\lfloor \frac{b - 2i}{2} \right\rfloor + (b - 2t - 1).$$

For $b = 2r$ this exceeds $b - 1$ by

$$\frac{t(2r - t - 3)}{2} \geq 0 \quad (0 \leq t \leq r - 1, r \geq 3),$$

and for $b = 2r + 1$ the same displayed expression is the excess over $b - 1$. Hence at least $b - 1$ good pairs are always counted. $\square$

Corollary 1. *The desired inequality holds for $p = 0$, for $p = 1/k$ with every integer $k \geq 3$, for $p = 2/b$ with every integer $b \geq 6$, for $p = 1/2$, and for $p = 1$.*

Proof. The cases $p = 0, 1$ are immediate. The reciprocal cases follow from Proposition 1 with $a = 1$, and Lemma 1 plus Proposition 1 gives $p = 2/b$, $b \geq 6$.

For $p = 1/2$, put $X = \sum_i w_i v_i$. Since $1 - v_i$ has the same law as v_i , X and $1 - X$ have the same distribution. Hence $\Pr[X < 1/2] = \Pr[X > 1/2]$, and so $\Pr[X \geq 1/2] \geq 1/2$. $\square$

A possible pointwise verification of the hypothesis of Proposition 1 would ask that, for every non-negative b -tuple of total 1, at least $\binom{b-1}{a-1}$ of the a -subsets have load at least a/b . This deterministic statement is false already at $a/b = 3/10$: take seven loads equal to $1/7$ and three loads equal to 0. A triple is then heavy only if all three of its points come from the seven positive positions, giving $\binom{7}{3} = 35 < 36 = \binom{9}{2}$. Thus the missing rational cases require genuinely averaged information about the random coloring, not just a pointwise Manickam–Miklos–Singhi type count.

Proposition 2 (multiplicative closure). *If $p, q \in \mathcal{G}$, then $pq \in \mathcal{G}$.*

Proof. If $pq = 0$, the conclusion is trivial. Otherwise generate independent Bernoulli(pq) variables as $v_i = x_i y_i$, where x_i are i.i.d. Bernoulli(p), y_i are i.i.d. Bernoulli(q), and the two families are independent. Put $T = \sum_i w_i x_i$. With probability at least p , $T \geq p$. Conditional on such an outcome, the normalized weights $w_i x_i / T$ form a probability vector, and the q -inequality gives

$$\Pr \left[\sum_i \frac{w_i x_i}{T} y_i \geq q \mid x_1, \dots, x_m \right] \geq q.$$

On this event $\sum_i w_i x_i y_i \geq qT \geq pq$, proving the claim. $\square$

Proposition 3 (closure of the set of good probabilities). *The set $\mathcal{G}$ is closed in $[0, 1]$.*

Proof. Let $p_n \in \mathcal{G}$ and $p_n \rightarrow p$. Fix a finite weight vector w . For $r \in [0, 1]$ put

$$F(r) = \Pr_r \left[\sum_i w_i v_i \geq r \right],$$

where the subscript denotes the Bernoulli parameter. There are only finitely many subset sums $w(A)$.

If infinitely many $p_n \leq p$, pass to that subsequence. Among the finitely many subset sums strictly smaller than p , let $p - \delta$ be the largest one, if it exists; otherwise take any $\delta > 0$. For all

large n , $p_n > p - \delta$, so no subset sum lies in $[p_n, p)$. Hence the event $\{w(A) \geq p_n\}$ is eventually the same family of subsets as $\{w(A) \geq p\}$. Therefore $F(p) = \lim F(p_n) \geq \lim p_n = p$.

If instead infinitely many $p_n > p$, pass to that subsequence. Let $p + \delta$ be the smallest subset sum strictly larger than p , if it exists; otherwise take any $\delta > 0$. For all large n , $p_n < p + \delta$, so the event $\{w(A) \geq p_n\}$ is the fixed family $\{w(A) > p\}$, and

$$\mu_p\{A : w(A) > p\} = \lim F(p_n) \geq p.$$

Since $\{w(A) > p\} \subseteq \{w(A) \geq p\}$, again $F(p) \geq p$. As the weight vector was arbitrary, $p \in \mathcal{G}$. $\square$

Corollary 2. *If the inequality can be proved for all $p \in [1/6, 1/3]$, then it holds for all $p \in [0, 1/3]$.*

Proof. For $0 < p \leq 1/3$, choose $n \geq 0$ such that $2^n p \in [1/6, 1/3]$. Since $1/2 \in \mathcal{G}$, repeated use of Proposition 2 gives $p = (2^n p)2^{-n} \in \mathcal{G}$. The point $p = 0$ is trivial. $\square$

Values that cannot work

Proposition 4. *The desired universal inequality fails for every $p \in (1/3, 1/2) \cup (1/2, 1)$.*

Proof. If $1/3 < p < 1/2$, take $m = 3$ and $w_1 = w_2 = w_3 = 1/3$. At least two successes are needed, so

$$\Pr\left[\frac{v_1 + v_2 + v_3}{3} \geq p\right] = 3p^2(1 - p) + p^3 = 3p^2 - 2p^3.$$

But $3p^2 - 2p^3 - p = p(1 - p)(2p - 1) < 0$ on $1/3 < p < 1/2$.

If $1/2 < p < 1$, take $m = 2$ and $w_1 = w_2 = 1/2$. The event requires two successes and has probability $p^2 < p$. $\square$

Equivalent forms of the remaining small- p assertion

Conjecture 1 (small- p assertion). *If $0 \leq p \leq 1/3$, $w_i \geq 0$, $\sum_i w_i = 1$, and the v_i 's are i.i.d. Bernoulli(p), then*

$$\Pr\left[\sum_i w_i v_i \geq p\right] \geq p.$$

The assertion is sharp even in finite examples: if one weight is at least p and the sum of all remaining weights is $< p$, then the event in (P) is exactly the event that this large coordinate succeeds, and its probability is p .

The nontrivial sharpness mechanism with all weights below p is a pivot plus dust. Fix $0 < p \leq 1/3$, choose $0 < \varepsilon < p$, put $a = p - \varepsilon$, and take

$$w_0 = a, \quad w_1 = \cdots = w_N = \frac{1 - a}{N}.$$

For all sufficiently large N , the dust weights $(1 - a)/N$, as well as the pivot a , are below p . If $Y_N = (1 - a)N^{-1}\text{Bin}(N, p)$, then

$$\Pr[w_0 v_0 + Y_N \geq p] = p \Pr[Y_N \geq \varepsilon] + (1 - p) \Pr[Y_N \geq p].$$

Since $Y_N \rightarrow p(1 - a) = p(1 - p + \varepsilon)$ in probability and $\varepsilon < p(1 - p + \varepsilon) < p$, this probability tends to p .

A particularly useful normalization is the following “unit-gap” form.

Proposition 5 (unit-gap form). *For a fixed $0 < p \leq 1/3$, Conjecture 1 is equivalent to the following assertion. Let $q = 1 - p$, let $0 \leq b_i \leq 1$, $B = \sum_i b_i$, and let ζ_i be i.i.d. Bernoulli(p). Then*

$$\Pr\left[\sum_i b_i \zeta_i \geq p(B + 1)\right] \geq \frac{p}{q} \Pr\left[\sum_i b_i \zeta_i < p(B + 1) - 1\right]. \quad (\text{UG})$$

Proof. Assume Conjecture 1. Apply it to the normalized coefficient vector $(1, b_1, \dots, b_n)/(B + 1)$, and condition on the Bernoulli variable multiplying the coefficient 1. With $S = \sum_i b_i \zeta_i$,

$$p \Pr[S \geq p(B + 1) - 1] + q \Pr[S \geq p(B + 1)] \geq p,$$

which rearranges to (UG).

Conversely, suppose (UG) is known. If an original weight is at least p , then that coordinate alone proves (P). Otherwise choose a pivot of maximal weight $a \in (0, p)$. Denote the residual weights by u_i , and write $u_i = ab_i$. Then $0 \leq b_i \leq 1$ and $a(B + 1) = 1$. Multiplying (UG) by a in the thresholds gives

$$\Pr\left[\sum_i u_i v_i \geq p\right] \geq \frac{p}{q} \Pr\left[\sum_i u_i v_i < p - a\right].$$

Conditioning on the pivot variable then yields

$$p \Pr\left[\sum_i u_i v_i \geq p - a\right] + q \Pr\left[\sum_i u_i v_i \geq p\right] \geq p,$$

which is (P). □

The same statement will sometimes be more convenient in the following homogeneous form. For every $h > 0$, every $0 \leq b_i \leq h$, and $B = \sum_i b_i$,

$$\Pr\left[\sum_i b_i \zeta_i \geq p(B + h)\right] \geq \frac{p}{q} \Pr\left[\sum_i b_i \zeta_i < p(B + h) - h\right]. \quad (\text{HUG})$$

Indeed, dividing all coefficients by h reduces (HUG) to (UG), and $h = 1$ recovers (UG). Thus Conjecture 1, (UG), and (HUG) are equivalent for each fixed $0 < p \leq 1/3$, with $p = 0$ trivial.

The following shifted version is one rigorous piece of the natural induction. It does not cover the pivot normalization needed for the full conjecture, but it identifies exactly where the one-coordinate induction breaks.

Proposition 6 (subcritical shifted unit-gap inequality). *Let $0 < p \leq 1/3$, $q = 1 - p$, $h > 0$, $0 \leq b_i \leq h$, $B = \sum_i b_i$, and $S = \sum_i b_i \zeta_i$, where the ζ_i 's are i.i.d. Bernoulli(p). For*

$$0 \leq \delta \leq qh - pB$$

one has

$$q \Pr[S \geq p(B + h) + \delta] \geq p \Pr[S < p(B + h) + \delta - h]. \quad (6)$$

In particular (HUG) is proved whenever $B \leq qh/p$.

Proof. We induct on the number of coefficients. If there are none, then $B = 0$, and for $0 \leq \delta \leq qh$ both events in (6) are empty (with the left threshold positive and the right threshold non-positive).

Write $S = R + c\zeta$, $B = B' + c$, with $0 \leq c \leq h$. Put $T = p(B + h) + \delta$. The difference between the two sides of (6) is

$$\begin{aligned} & q \Pr[S \geq T] - p \Pr[S < T - h] \\ &= q \left(q \Pr[R \geq T] - p \Pr[R < T - h] \right) + p \left(q \Pr[R \geq T - c] - p \Pr[R < T - h - c] \right). \end{aligned} \quad (7)$$

For the first bracket, observe that

$$T = p(B' + h) + (\delta + pc),$$

and

$$0 \leq \delta + pc \leq qh - pB + pc = qh - pB'.$$

Thus the induction hypothesis applies to R with shift $\delta + pc$.

For the second bracket,

$$T - c = p(B' + h) + (\delta - qc).$$

If $\delta - qc \geq 0$, then

$$\delta - qc \leq \delta \leq qh - pB \leq qh - pB',$$

so induction applies again. If $\delta - qc < 0$, then $T - c < p(B' + h)$ and $T - h - c < p(B' + h) - h$. Hence

$$\Pr[R \geq T - c] \geq \Pr[R \geq p(B' + h)]$$

and

$$\Pr[R < T - h - c] \leq \Pr[R < p(B' + h) - h].$$

The induction hypothesis at shift 0 therefore makes the second bracket non-negative. Both brackets in (7) are non-negative, completing the induction. $\square$

For rational parameters there is also a useful exact cyclic formulation. The case $p = 0$ is trivial, so let $p = r/b \leq 1/3$ with $1 \leq r \leq b$, in lowest or non-lowest terms. Assign each

coefficient w_i independently and uniformly to one of the b cyclic positions $\mathbb{Z}/b\mathbb{Z}$, and let L_j be the total load in the cyclic interval $[j, j+r)$. Then

$$\Pr\left[\sum_i w_i v_i < p\right] = \frac{1}{b} \mathbb{E} \#\{j : L_j < r/b\}. \quad (8)$$

Consequently the rational case of Conjecture 1 is equivalent to the averaged cyclic inequality

$$\mathbb{E} \#\{j : L_j < r/b\} \leq b - r. \quad (9)$$

The corresponding deterministic assertion for each fixed load vector is false. For $r/b = 2/7$, the cyclic load vector

$$\left(\frac{1}{4}, \frac{1}{4}, 0, \frac{1}{4}, 0, \frac{1}{4}, 0\right)$$

has six of its seven adjacent 2-intervals of load $1/4 < 2/7$, and only the interval through the first two positions is heavy. Thus a cyclic proof would have to use the randomness of the allocation, not just a pointwise interval count.

There is also a useful sequential identity for $0 < p \leq 1/3$. Fix any ordering of the original weights and put $S_i = \sum_{j \leq i} w_j v_j$, $S_0 = 0$. Since the increments are non-negative,

$$\Pr[S_i < p] = \Pr[S_{i-1} < p] - p \Pr[p - w_i \leq S_{i-1} < p].$$

Iterating gives

$$\Pr[S_m < p] = 1 - p \sum_{i=1}^m \Pr[p - w_i \leq S_{i-1} < p]. \quad (10)$$

Hence the small- p assertion is equivalent to the crossing-intensity bound

$$\sum_{i=1}^m \Pr[p - w_i \leq S_{i-1} < p] \geq 1$$

for every deterministic ordering of the weights.

Here are exact reformulations of Conjecture 1.

First, after replacing v_i by $1 - v_i$, it is equivalent to

$$q \geq \frac{2}{3}, \quad \eta_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(q) \quad \implies \quad \Pr\left[\sum_i w_i \eta_i > q\right] \leq q, \quad q = 1 - p. \quad (1)$$

The inequality is strict on the left because $\sum_i w_i v_i \geq p$ is the same as $\sum_i w_i (1 - v_i) \leq 1 - p$.

Second, it is equivalent to the following two-tail pivot inequality: for every $0 \leq a \leq p$, every set of weights $u_i \geq 0$ with $\sum_i u_i = 1 - a$, and i.i.d. $v_i \sim \text{Bernoulli}(p)$,

$$\Pr\left[\sum_i u_i v_i \geq p\right] \geq \frac{p}{1-p} \Pr\left[\sum_i u_i v_i < p - a\right]. \quad (2)$$

Indeed, (2) with $a = 0$ gives $U \geq \frac{p}{1-p}(1 - U)$, hence $U \geq p$. Conversely, if Conjecture 1 is known, add one extra weight a . Conditioning on its Bernoulli variable gives

$$p \Pr[Y \geq p - a] + (1 - p) \Pr[Y \geq p] \geq p,$$

where $Y = \sum_i u_i v_i$; rearranging gives (2).

Third, let μ_p denote p -biased product measure on $2^{[m]}$. The small- p assertion is equivalent to the following convex-hull statement: for every non-empty down-set $\mathcal{D} \subseteq 2^{[m]}$,

$$\mu_p(\mathcal{D}) > 1 - p \implies (p, \dots, p) \in \text{conv}\{\mathbf{1}_A : A \in \mathcal{D}\}. \quad (3)$$

For the forward implication, take $\mathcal{D} = \{A : \sum_{i \in A} w_i < p\}$. If (3) supplied a distribution on $\mathcal{D}$ with all coordinate marginals p , then

$$p = \sum_i p w_i = \mathbb{E} \sum_{i \in A} w_i < p,$$

a contradiction.

Conversely, suppose (3) fails for a down-set $\mathcal{D}$, and put $P(\mathcal{D}) = \text{conv}\{\mathbf{1}_A : A \in \mathcal{D}\}$. This convex set is coordinatewise down-closed: if $x \in P(\mathcal{D})$ and $0 \leq y \leq x$, represent x as the marginals of a random $A \in \mathcal{D}$, then thin each $i \in A$ independently with probability y_i/x_i (with the obvious convention when $x_i = 0$). The resulting random subset is still in $\mathcal{D}$ and has marginals y . Thus $P(\mathcal{D})$ is disjoint from the closed upper orthant $(p, \dots, p) + \mathbb{R}_{\geq 0}^m$. Since $P(\mathcal{D})$ is compact and the two closed convex sets have positive distance, strict separation gives a non-zero normal vector c . The vector must have $c_i \geq 0$ for each i : otherwise the linear functional $x \mapsto c \cdot x$ would be unbounded below on the upper orthant $(p, \dots, p) + \mathbb{R}_{\geq 0}^m$ and could not separate it from the compact polytope $P(\mathcal{D})$. Thus

$$\sum_{i \in A} c_i < p \sum_i c_i \quad (A \in \mathcal{D})$$

for some non-zero $c \geq 0$. After normalizing $w_i = c_i / \sum_j c_j$, Conjecture 1 would imply

$$\mu_p(\mathcal{D}) \leq \Pr \left[\sum_{i \in A} w_i < p \right] \leq 1 - p,$$

contradicting $\mu_p(\mathcal{D}) > 1 - p$.

Equivalently, with $\lambda = p/(1 - p)$, the remaining down-set statement is the partition-function bound

$$p\mathbf{1} \notin P(\mathcal{D}) \implies \sum_{A \in \mathcal{D}} \lambda^{|A|} \leq (1 + \lambda)^{m-1}, \quad 0 \leq \lambda \leq \frac{1}{2}. \quad (4)$$

The following slice formulation records the sharpest induction target I have isolated. If

$$\mathcal{D} = \mathcal{E} \cup \{A \cup \{n + 1\} : A \in \mathcal{F}\}, \quad \mathcal{F} \subseteq \mathcal{E} \subseteq 2^{[n]},$$

then $Z_\lambda(\mathcal{D}) = Z_\lambda(\mathcal{E}) + \lambda Z_\lambda(\mathcal{F})$. For $c \geq 0$, put

$$r_{\mathcal{F}}(c) = \max_{A \in \mathcal{F}} c(A), \quad \rho_{\mathcal{E}}^\lambda(c) = \max\{c \cdot x : x \in P(\mathcal{E}), 0 \leq x_i \leq \lambda\}.$$

The condition $p\mathbf{1} \notin P(\mathcal{D})$ separates the down-convex set $P(\mathcal{F})$ from the reflected capped set $\{y : \lambda(\mathbf{1} - y) \in P(\mathcal{E})\}$. Thus the missing induction would follow from the following capped replacement lemma:

$$\rho_{\mathcal{E}}^\lambda(c) + \lambda r_{\mathcal{F}}(c) < \lambda c([n]) \implies Z_\lambda(\mathcal{E}) + \lambda Z_\lambda(\mathcal{F}) \leq (1 + \lambda)^n.$$

This lemma is not proved here; it is included to make the remaining obstruction precise. The uncapped scalar diagonal reach loses information in exactly the mixed-slice case $p\mathbf{1} \in P(\mathcal{E})$ but $p\mathbf{1} \notin P(\mathcal{F})$.

Finally, for $0 < p \leq 1/3$, the same down-set has a useful intersection-theoretic description. If $\mathcal{D}_w = \{A : \sum_{i \in A} w_i < p\}$ and $r = \lfloor 1/p \rfloor$, then no r members of $\mathcal{D}_w$ cover $[m]$, because the weight of their union is $< rp \leq 1$. Equivalently, the complement family

$$\mathcal{H}_w = \{B : \sum_{i \in B} w_i > 1 - p\}$$

is r -wise intersecting. Conjecture 1 is therefore the special weighted-threshold assertion that such a family has $(1 - p)$ -biased measure at most $1 - p$.

A coalescence identity and a failed route

The complement form also gives a useful local identity. Put $q = 1 - p$ and, for a probability vector w , define

$$\Phi_q(w) = \Pr \left[\sum_i w_i \eta_i > q \right], \quad \eta_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(q).$$

Thus Conjecture 1 is equivalent to $\Phi_q(w) \leq q$ for all $q \geq 2/3$.

If two coordinates of weights $x \leq y$ are replaced by one coordinate of weight $x + y$, and Z denotes the contribution of all other coordinates, then direct conditioning on Z gives

$$\begin{aligned} & \Phi_q(\dots, x + y, \dots) - \Phi_q(\dots, x, y, \dots) \\ &= pq \left(\Pr[q - x - y < Z \leq q - y] - \Pr[q - x < Z \leq q] \right), \end{aligned} \tag{5}$$

with the displayed half-open endpoints corresponding to the strict event $\sum w_i \eta_i > q$.

A tempting induction would be to prove that, unless only one coordinate remains, some pair has non-negative gain in (5). This is false, even in the capped regime $w_i < p$. For example, take

$$p = \frac{3}{10}, \quad q = \frac{7}{10}, \quad w_1 = w_2 = w_3 = w_4 = \frac{1}{4}.$$

Then

$$\Phi_q(w) = \Pr[\text{Bin}(4, q) \geq 3] = 4q^3(1 - q) + q^4 = \frac{6517}{10000}.$$

After coalescing any pair the vector is $(1/2, 1/4, 1/4)$, and the event

$$\frac{1}{2}\eta_0 + \frac{1}{4}\eta_1 + \frac{1}{4}\eta_2 > 0.7$$

occurs exactly when the merged coordinate succeeds and at least one of the two remaining coordinates succeeds. Hence the new value is

$$q(1 - (1 - q)^2) = \frac{637}{1000},$$

which is smaller. Thus pairwise coalescence cannot by itself prove Conjecture 1.

The obstruction is not isolated. If $n \geq 4$, $1/n < p < 1/(n - 1)$, and $w_i = 1/n$, then $w_i < p$ and

$$\Phi_q(w) = q^n + npq^{n-1},$$

because the p -biased failure set has weight $< p$ exactly when it is empty or a singleton. Coalescing any pair gives

$$\Phi_q(w^{ij}) = q^{n-1} + (n - 2)pq^{n-2},$$

so

$$\Phi_q(w^{ij}) - \Phi_q(w) = pq^{n-2}(n - 2 - (n - 1)q) < 0.$$

Any successful coalescence proof must therefore use larger or adaptive blocks, or combine coalescence with a separate treatment of these reciprocal-chamber configurations.

Remaining open issues

The negative direction is complete, and the positive direction is complete for the values listed in Corollary 1, their finite products under Proposition 2, and $p = 1/2, 1$. Therefore a full solution would follow from Conjecture 1; combined with Proposition 4, it would give exactly $[0, 1/3] \cup \{1/2, 1\}$.

The essential missing assertion is Conjecture 1. Mathematically, it is the assertion that every weighted threshold down-set $\{A : \sum_{i \in A} w_i < p\}$ has p -biased measure at most $1 - p$ when $p \leq 1/3$. It appears in the document as the unproved implication $[0, 1/3] \subseteq \mathcal{G}$. The approaches recorded above reduce it to any one of the unit-gap inequality (UG), the strict complement inequality (1), the pivot inequality (2), the convex-hull statement (3), or the partition-function inequality (4). Proposition 3 shows that it would also be enough to prove the inequality on any dense subset of $[0, 1/3]$, or on the base interval $[1/6, 1/3]$ by Corollary 2.

The pairwise coalescence assertion suggested by formula (5) is not available: the equal-weight examples in the preceding section show that every pair can have negative coalescence gain. A viable coalescence argument would need either block coalescence or a dichotomy proving that configurations with all pair gains negative already obey $\Phi_q(w) \leq q$.

The capped slice lemma displayed after (4) is the most focused remaining inductive target. Exhaustive enumeration of all down-sets on at most five coordinates found no violation of the convex-hull statement (3), and the extremal separated examples were only the expected dictatorship equalities and lower-measure singleton-cone examples. These computations are evidence for (3), but of course not a proof.

The complement form is also adjacent to the hypotenuse case $t = q$ of the “bold play” problem studied by Fokkink–Meester–Pelekis [1]: maximize $\Pr(\sum c_i \beta_i \geq t)$ over convex combinations of iid Bernoulli(q) variables. For a fixed finite weight vector the strict event $\sum c_i \beta_i > q$ is the same as $\sum c_i \beta_i \geq t_*$, where t_* is the next attainable subset sum above q . Their published results prove several adjacent rectangles and reciprocal boundary points, but the near-hypotenuse triangles with $q < t_*$ close to q , needed for all $q \geq 2/3$, are not derived here from those results. Frankl-type intersection theorems [2] explain some of the surrounding extremal combinatorics, but they do not presently supply the weighted-threshold biased measure bound (4).

References

- [1] R. Fokkink, L. Meester, and C. Pelekis, *Optimizing stakes in simultaneous bets*, ALEA Lat. Am. J. Probab. Math. Stat. 20 (2023), 153–165.
- [2] P. Frankl, *On Sperner families satisfying an additional condition*, J. Combin. Theory Ser. A 20 (1976), 1–11.